

AUTO LICENSE— FULL TECHNICAL REFERENCE v2.0

1. SYSTEM OVERVIEW

Platform: Mobile (Flutter) + Backend (REST API)

Users: Driver, Company, Admin

Purpose: Taxi driver license lifecycle automation

2. ARCHITECTURE

Layers:

- Presentation: Flutter UI, Widgets
- State: BLoC / Cubit
- Data: Repository pattern
- Network: REST API (Dio)
- Storage: SecureStorage (tokens), local cache
- Realtime: Push (FCM)

3. AUTH FLOW

1. Phone input (+998XXXXXXXXXX)
2. OTP send → /auth/otp/send
3. OTP verify → /auth/otp/verify
4. Response: access_token + refresh_token
5. Store: secure storage
6. Auto refresh on 401

4. MAIN MODULES

4.1 USER

Fields:

- id (UUID)
- phone
- role
- created_at

4.2 DRIVER PROFILE

- full_name
- car_info
- license_id

4.3 LICENSE

- id
- driver_id
- license_number
- expiry_date
- status: Active / Expiring / Expired

Logic:

expiry_date - today <= 15 → Expiring

4.4 PAYMENT

- id
- user_id
- amount
- method
- status: pending / success / failed

4.5 NOTIFICATION

Triggers:

- 15 days before expiry
- 7 days before expiry

Channels:

- Push (FCM)
- SMS

5. BUSINESS LOGIC

IF expiry_date - 15:

→ send notification

IF user clicks "Continue":

→ open payment

IF payment success:

→ create new license

→ update expiry_date

IF user cancels:

→ stop process

6. API STRUCTURE

Auth:

POST /auth/otp/send
POST /auth/otp/verify
POST /auth/refresh

User:

GET /users/me

License:

POST /licenses
GET /licenses/:id
PATCH /licenses/:id

Payment:

POST /payments
GET /payments/:id

Notification:

GET /notifications

7. GAI VERIFICATION

Methods:

- QR code scan
- License ID search

Returns:

- Driver name
- License number
- Expiry date
- Status

8. DATABASE STRUCTURE

Users(id, phone, role)

Drivers(user_id, full_name, car_info)

Licenses(id, driver_id, number, expiry, status)

Payments(id, user_id, amount, status)

9. SECURITY

- JWT (access 15m, refresh 30d)
- HTTPS
- Role-based access
- OTP verification

10. PERFORMANCE

- Response time $\leq 500\text{ms}$
- Uptime $\geq 99\%$
- Scalable (100k users)

11. MVP SCOPE

- Auth
- License module
- Payment
- Notification
- QR verification

12. FUTURE

- Government API integration
- Auto-renewal
- Analytics dashboard

MVP-MAP:

Category	Details
Auth	Phone + OTP, JWT
Driver Profile	FIO, Car info
License	Add license, expiry_date, status: Active/Expired
Notification	7 days before expiry
Payment	1 provider (Click/Payme)
Electronic License	Simple view + QR
GAI Check	GET /license/:id
Database	Users, Drivers, Licenses, Payments
Performance	<=10k users, ~1s response

PRODUCTION-MAP:

Category	Details
Auth	OTP + JWT + Refresh Token
Roles	Driver, Company, Admin
License Lifecycle	Active → Expiring → Expired → Renewed
Notification	15 & 7 days, Push + SMS, Retry
Payment	Click, Payme, Uzcard, Humo + Webhook
Electronic License	PDF + QR + Unique ID
GAI Check	Public API + QR Scan
Admin Panel	Users, Licenses, Payments, Stats
B2B Panel	Driver management
Security	JWT, HTTPS, RBAC, Rate limit
Performance	100k+ users, <=500ms
Realtime	Push, Queue, Retry
Edge Cases	Idempotency, retry, fallback